

5.1 Energy: Conservation, Work, Power & Efficiency

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.1 Energy: Conservation, Work, Power & Efficiency
Difficulty	Easy

Time allowed: 30

Score: /25

Percentage: /100

Question 1a

A small child does work on a large suitcase by pushing it along a smooth floor, as shown in Fig. 1.1.



Fig. 1.1.

State the definition of work done and write the equation.

[2 marks]

Question 1b

The suitcase is moved back to its original position by an adult who pulls it at an angle of 70° to the floor, as shown in Fig. 1.2.

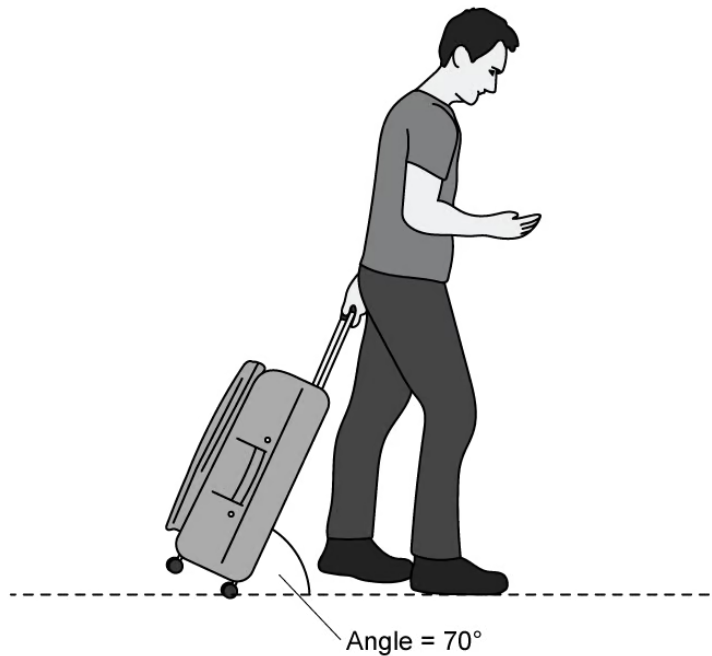


Fig. 1.2.

Write an expression for the work done on the suitcase by the adult in terms of force F , distance d and the angle 70° .

[2 marks]

Question 1c

In moving the suitcase the child and adult both exerted equal forces of 20 N. They each move the suitcase a distance of 2.5 m.

Calculate the work done by

(i)
the child

[2]

(ii)
the adult.

[2]

[4 marks]

Question 1d

State whether the child or the adult are more efficient at moving the suitcase, explaining your reasoning.

[3 marks]

Question 2a

A block of weight $4.25 \times 10^3 \text{ N}$ is placed on a frictionless slope inclined at 30° to the horizontal as shown in Fig. 1.1.

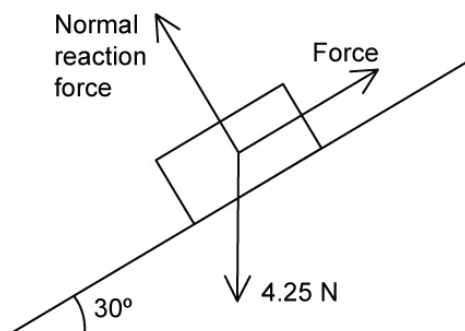


Fig. 1.1.

The block is moved up the slope at constant speed by applying a force parallel to the slope.

Calculate the component of the weight of the block which acts parallel to the slope.

[2 marks]

Question 2b

Calculate the work done in moving the barrel a distance of 6.0 m up the slope.

[2 marks]

Question 2c

The block in part (a) is pushed 6.0 m up the slope in 18 s.

Calculate the power needed to move the block in this amount of time.

[2 marks]

Question 3a

A beaker is filled with ice and then an electric heater placed into it, as shown in Fig. 1.1. The heater is turned on and heats up until it glows red.

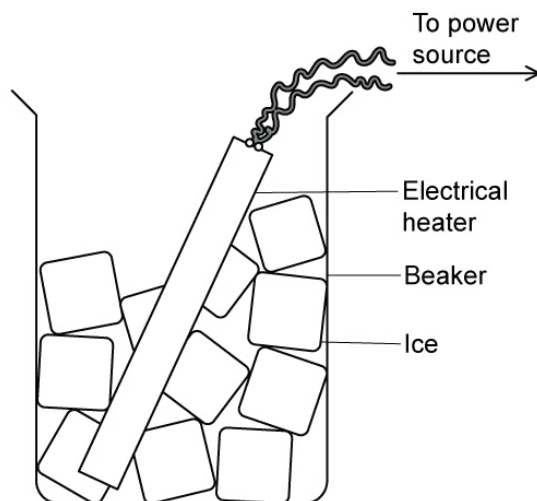


Fig. 1.1.

A student states that when the ice is being heated up the total amount of energy in the ice remains the same.

State the principle of conservation of energy.

[2 marks]

Question 3b

Explain whether the student is correct, giving your reason.

[3 marks]

Question 3c

Identify the energy transfers involved in heating the ice by filling in the missing words below.

_____ energy $\longrightarrow$ _____ energy + _____ energy

[3 marks]

Model Answers: Easy

1a

State the definition of work done and write the equation.:

- The energy needed to move an object over a distance by an external force applied in the direction of its displacement; [1 mark]
- $W = F \times d$; [1 mark]

[Total: 2 marks]

1b

Write an equation which would be used to calculate the work done on the suitcase by the adult:

- Work done on an incline: $W = F \times d \cos \theta$ [1 mark]
- Where $F \cos \theta$ = the component of the force (in the direction of the motion) and d = horizontal distance moved
- $W = F \times d \cos 70$ [1 mark]

[Total: 2 marks]

1c

i) Calculate the work done by the child:

Substitute the values and calculate:

- $W = 20 \times 2.5$ [1 mark]
- $W = 50 \text{ J}$ [1 mark]

ii) Calculate the work done by the adult:

- $W = (20 \times \cos 70) \times 2.5$ [1 mark]

- $W = 17 \text{ J}$ [1 mark]

[Total: 4 marks]

You must have correct values **and** units to get the two 'final answer' marks.

1d

State whether the child or the adult are more efficient at moving the suitcase, explaining your reasoning:

Statement

- The adult is more efficient: [1 mark]

Reason

- They used less energy to move the case the same distance; [1 mark]

Explanation

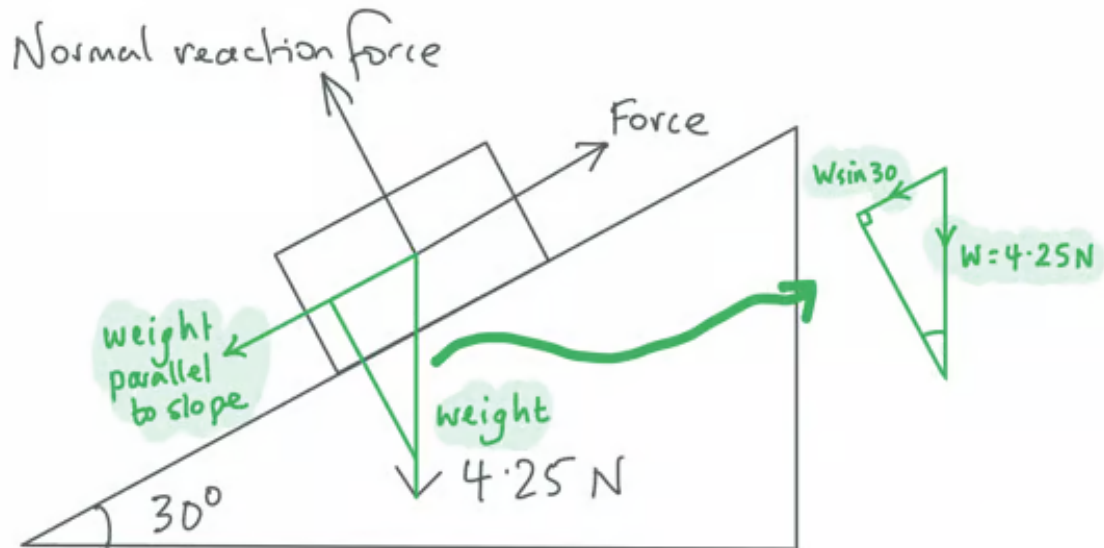
- The angle means less force in the direction of motion **OR** only two wheels / less wheels on the floor / less friction with the floor; [1 mark]

[Total: 3 marks]

2a

Calculate the component of the weight of the block which acts parallel to the slope:

Draw a vector triangle to identify the correct component:



Calculate for the component of weight parallel to slope:

- $W_{\text{slope}} = W \sin 30 = 4.25 \times \sin 30$ [1 mark]
- $W_{\text{slope}} = 2.1 \text{ N (2 s.f.)}$ [1 mark]

[Total: 2 marks]

2b

Calculate the work done in moving the barrel a distance of 6.0 m up the slope:

Use the equation for work done:

- $W = F \times d$ where F = force in the direction of the displacement
- $W = 2.1 \times 6.0$ [1 mark]
- $W = 13 \text{ N or } 12.6 \text{ N}$ [1 mark]

[Total: 2 marks]

2c

Calculate the power needed to move the block in this amount of time:

Use the equation for power:

- $P = \frac{W}{t} = \frac{13}{18}$ **or** $\frac{12.6}{18}$ [1 mark]
- $P = 0.72 \text{ W}$ **or** 0.70 W (2 s.f.) [1 mark]

[Total: 2 marks]

Notice that to quote 0.7 to 2 significant figures you need to include the final zero.

3a

State the principle of conservation of energy:

- Energy cannot be created or destroyed; [1 mark]
- It can only change from one form to another; [1 mark]

[Total: 2 marks]

3b

Explain whether the student is correct, giving your reason:

- The student is not correct; [1 mark]
- (Conservation of energy means the) energy in a closed system remains constant; [1 mark]
- Heating ice is not closed / heat escapes (from the system / from the heater / through convection / conduction); [1 mark]

[Total: 3 marks]

3c

Identify the energy transfers involved:

- Heat and light can be in either order:

electrical energy [1 mark] → **heat** energy [1 mark] → **light** energy [1 mark]

[Total: 3 marks]